

5. HEALTH CARE ANALYTICS SYSTEM:

Introduction:

A Health Care Analytics System Manages large amounts of data from Hospitals, Laboratories, Electronic Health Records (EHRs), Medical devices and wearable devices. Big data Technologies help store process and analyze this information to improve patients care, predict diseases and supports doctors in Decision Making.

1. Hybrid Storage Architecture

Healthcare data can be structured, semi-structured or unstructured. Patient Records, appointments, prescriptions and laboratory results can be stored in relational database such as MySQL or PostgreSQL. Medical Images, doctors notes, sensor data and other large files can be stored using NoSQL, databases, HDFS or cloud storage.

A data lake can store raw health care data, while a data warehouse can store cleaned data for reporting and analytics.

2. ETL and Data Integration

Data is collected from different sources and processed using an ETL Pipeline

Data Collection → cleaning → Transformation
→ Storage → Analysis

ETL removes Duplicate or incorrect data and converts information into a common format. Health care standard such as HL7 and FHIR help different Health care systems exchange patient information efficiently.

3. Real-time Monitoring

Wearable devices can continuously collect information such as heart rate, Temperature and activity. Tools such as Apache, Kafka, Spark Streaming, or Flink can process this data in Real time.

If abnormal readings are detected, The system can immediately generate an alert for doctors or medical staff. This helps in early detection and faster response.

4. Distributed Processing

Healthcare organisations generate very large datasets. So, distributed processing is required. Apache Spark can divide data processing across multiple computers, making analysis faster and more scalable.

It can be used for:

- Patient data analysis
- Historical data processing
- * Machine - Learning Model Training
- * Population health analysis

5. Machine Learning Risk and Prediction

Machine Learning can identify patterns in patient data and support healthcare decisions. It can be used for

- * Disease risk prediction
- * Patient deterioration prediction
- * Readmission Prediction
- * Diagnosis risk Scoring
- * Treatment Recommendation

For Example:

Patient History + vital signs + Lab results $\rightarrow$ ML Model $\rightarrow$ Risk score

Patients can be classified into low-medium or high risk groups. ML recommendations should support doctors rather than replace professional Medical Judgement.

6. Data security and Governance

Healthcare Information is sensitive and must be protected. The system should use role-based access control, authentication, encryption and audit logs.

Access should be limited to authorized users such as Doctors, Nurses and administrators. Encryption protects information during storage and transmission. While audit logs records who accessed the data and when.

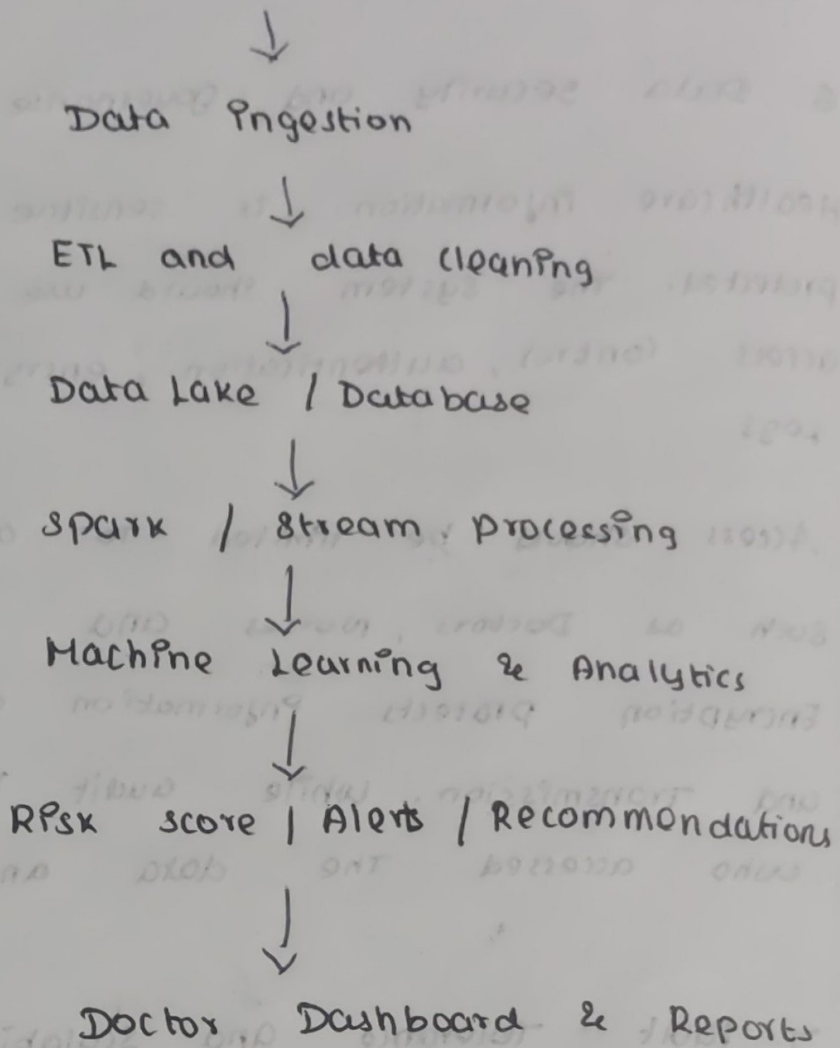
7. Fault Tolerance and Scalability

Health care system must remain available even when failures occur. Data Replication, Backups, load Balancing and failover Mechanisms improve reliability.

Cloud Infrastructure can also provide scalability by increasing computing and storage resources when the amount of Healthcare data grows.

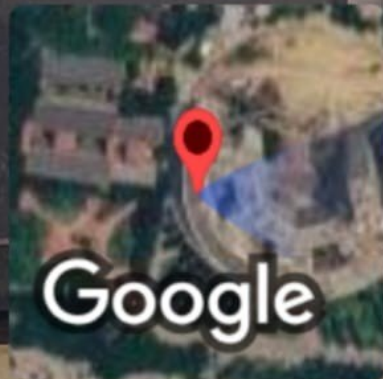
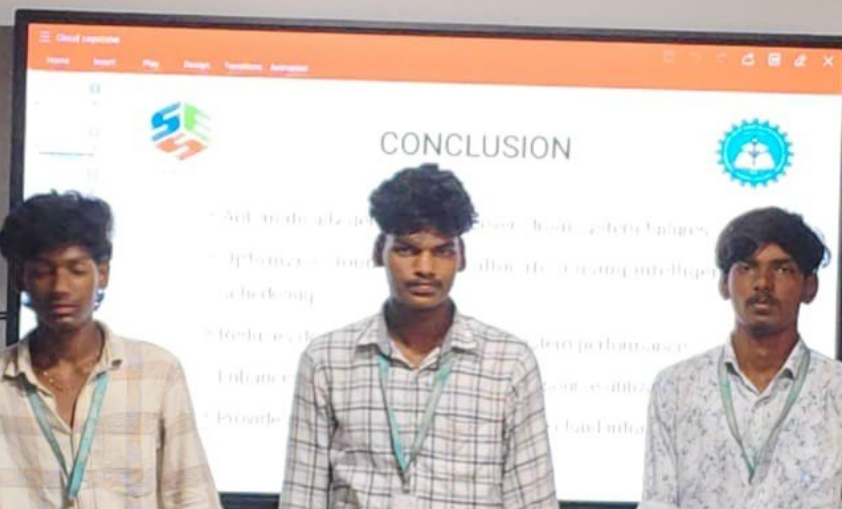
Overall Architecture:

Hospitals, Laboratories, EHRs, Wearables



Conclusion:

A Health care Analytics system combines Big data storage, ETL, real-time processing, distributed computing and Machine learning to manage health care information effectively. It enables realtime patient monitoring, risk prediction and better clinical decision making while maintaining data security, Reliability and scalability.



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